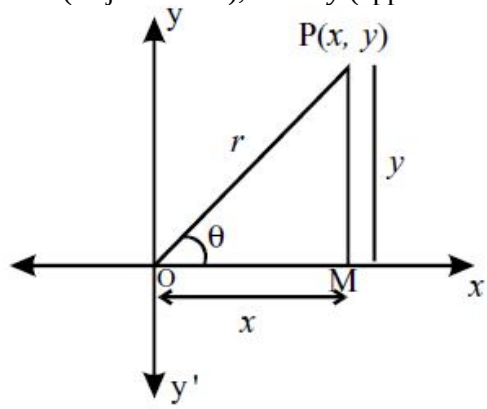
CLASS X: CHAPTER - 8
INTRODUCTION TO TRIGONOMETRY

IMPORTANT FORMULAS & CONCEPTS

The word 'trigonometry' is derived from the Greek words 'tri' (meaning three), 'gon' (meaning sides) and 'metron' (meaning measure). In fact, **trigonometry** is the study of relationships between the sides and angles of a triangle.

Trigonometric Ratios (T - Ratios) of an acute angle of a right triangle

In XOY-plane, let a revolving line OP starting from OX, trace out $\angle XOP=\theta$.
From P (x, y)draw PM $\perp$ to OX.
In right angled triangle OMP. OM = x (Adjacent side); PM = y (opposite side); OP = r (hypotenuse).



$$\sin\theta = \frac{\text{Opposite side}}{\text{Hypotenuse}} = \frac{y}{r} \qquad \cos ec\theta = \frac{\text{Hypotenuse}}{\text{Opposite side}} = \frac{r}{y}$$

$$\cos\theta = \frac{\text{Adjacent Side}}{\text{Hypotenuse}} = \frac{x}{r} \qquad \sec\theta = \frac{\text{Hypotenuse}}{\text{Adjacent Side}} = \frac{r}{x}$$

$$\tan\theta = \frac{\text{Opposite side}}{\text{Adjacent Side}} = \frac{y}{x} \qquad \cot\theta = \frac{\text{Adjacent Side}}{\text{Opposite side}} = \frac{x}{y}$$

Reciprocal Relations

$$\sin\theta = \frac{1}{\cos ec\theta} \qquad \cos ec\theta = \frac{1}{\sin\theta}$$

$$\cos\theta = \frac{1}{\sec\theta} \qquad \sec\theta = \frac{1}{\cos\theta}$$

$$\tan\theta = \frac{1}{\cot\theta} \qquad \cot\theta = \frac{1}{\tan\theta}$$

Quotient Relations

$$\tan\theta = \frac{\sin\theta}{\cos\theta} \quad \textit{and} \quad \cot\theta = \frac{\cos\theta}{\sin\theta}$$

- **Remark 1 :** sin q is read as the "sine of angle q" and it should never be interpreted as the product of 'sin' and 'q'
- **Remark 2 : Notation :** $(\sin\theta)^2$ is written as $\sin^2\theta$ (read "sin square q'') Similarly $(\sin\theta)^n$ is written as $\sin^n\theta$ (read ''sin *n*th power q''), *n* being a positive integer.
- **Note :** $(\sin\theta)^2$ should not be written as $\sin\theta^2$ or as $\sin^2\theta^2$
- **Remark 3 :** Trigonometric ratios depend only on the value of θ and are independent of the lengths of the sides of the right angled triangle.

Trigonometric ratios of Complementary angles.

$\sin(90 - \theta) = \cos\theta$ $\quad\quad$ $\cos(90 - \theta) = \sin\theta$

$\tan(90 - \theta) = \cot\theta$ $\quad\quad$ $\cot(90 - \theta) = \tan\theta$

$\sec(90 - \theta) = \operatorname{cosec}\theta$ $\quad\quad$ $\operatorname{cosec}(90 - \theta) = \sec\theta$.

Trigonometric ratios for angle of measure.

0^0, 30^0,45^0, 60^0 and 90^0 in tabular form.

$\angle$A	0^0	30^0	45^0	60^0	90^0
sinA	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cosA	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tanA	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	Not defined
cosecA	Not defined	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1
secA	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	Not defined
cotA	Not defined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0

TRIGONOMETRIC IDENTITIES

An equation involving trigonometric ratios of an angle is said to be a trigonometric identity if it is satisfied for all values of θ for which the given trigonometric ratios are defined.

Identity (1) : $\sin^2\theta + \cos^2\theta = 1$

$\Rightarrow \sin^2\theta = 1 - \cos^2\theta$ and $\cos^2\theta = 1 - \sin^2\theta$.

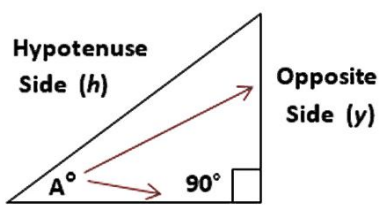
Identity (2) : $\sec^2\theta = 1 + \tan^2\theta$

$\Rightarrow \sec^2\theta - \tan^2\theta = 1$ and $\tan^2\theta = \sec^2\theta - 1$.

Identity (3) : $\operatorname{cosec}^2\theta = 1 + \cot^2\theta$

$\Rightarrow \operatorname{cosec}^2\theta - \cot^2\theta = 1$ and $\cot^2\theta = \operatorname{cosec}^2\theta - 1$.

SOME TIPS

Right Triangle	SOH-CAH-TOA Method	Coordinate System Method
	**SOH:** $\text{sine}(A) = \textbf{sin}(A) = \frac{\text{Opposite}}{\text{Hypotenuse}}$ **CAH:** $\text{cosine}(A) = \textbf{cos}(A) = \frac{\text{Adjacent}}{\text{Hypotenuse}}$ **TOA:** $\text{tangent}(A) = \textbf{tan}(A) = \frac{\text{Opposite}}{\text{Adjacent}}$ $\text{cosecant}(A) = \textbf{csc}(A) = \frac{1}{\sin(A)} = \frac{\text{Hypotenuse}}{\text{Opposite}}$ $\text{secant}(A) = \textbf{sec}(A) = \frac{1}{\cos(A)} = \frac{\text{Hypotenuse}}{\text{Adjacent}}$ $\text{cotangent}(A) = \textbf{cot}(A) = \frac{1}{\tan(A)} = \frac{\text{Adjacent}}{\text{Opposite}}$	$\textbf{sin}(A) = \frac{y}{h}$ $\textbf{cos}(A) = \frac{x}{h}$ $\textbf{tan}(A) = \frac{y}{x}$ $\textbf{csc}(A) = \frac{1}{\sin(A)} = \frac{h}{y}$ $\textbf{sec}(A) = \frac{1}{\cos(A)} = \frac{h}{x}$ $\textbf{cot}(A) = \frac{1}{\tan(A)} = \frac{x}{y}$

Each trigonometric function in terms of the other five.

in terms of	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
$\sin\theta =$	$\sin\theta$	$\pm\sqrt{1-\cos^2\theta}$	$\pm\frac{\tan\theta}{\sqrt{1+\tan^2\theta}}$	$\frac{1}{\csc\theta}$	$\pm\frac{\sqrt{\sec^2\theta-1}}{\sec\theta}$	$\pm\frac{1}{\sqrt{1+\cot^2\theta}}$
$\cos\theta =$	$\pm\sqrt{1-\sin^2\theta}$	$\cos\theta$	$\pm\frac{1}{\sqrt{1+\tan^2\theta}}$	$\pm\frac{\sqrt{\csc^2\theta-1}}{\csc\theta}$	$\frac{1}{\sec\theta}$	$\pm\frac{\cot\theta}{\sqrt{1+\cot^2\theta}}$
$\tan\theta =$	$\pm\frac{\sin\theta}{\sqrt{1-\sin^2\theta}}$	$\pm\frac{\sqrt{1-\cos^2\theta}}{\cos\theta}$	$\tan\theta$	$\pm\frac{1}{\sqrt{\csc^2\theta-1}}$	$\pm\sqrt{\sec^2\theta-1}$	$\frac{1}{\cot\theta}$
$\csc\theta =$	$\frac{1}{\sin\theta}$	$\pm\frac{1}{\sqrt{1-\cos^2\theta}}$	$\pm\frac{\sqrt{1+\tan^2\theta}}{\tan\theta}$	$\csc\theta$	$\pm\frac{\sec\theta}{\sqrt{\sec^2\theta-1}}$	$\pm\sqrt{1+\cot^2\theta}$
$\sec\theta =$	$\pm\frac{1}{\sqrt{1-\sin^2\theta}}$	$\frac{1}{\cos\theta}$	$\pm\sqrt{1+\tan^2\theta}$	$\pm\frac{\csc\theta}{\sqrt{\csc^2\theta-1}}$	$\sec\theta$	$\pm\frac{\sqrt{1+\cot^2\theta}}{\cot\theta}$
$\cot\theta =$	$\pm\frac{\sqrt{1-\sin^2\theta}}{\sin\theta}$	$\pm\frac{\cos\theta}{\sqrt{1-\cos^2\theta}}$	$\frac{1}{\tan\theta}$	$\pm\sqrt{\csc^2\theta-1}$	$\pm\frac{1}{\sqrt{\sec^2\theta-1}}$	$\cot\theta$

Note: $\csc\theta$ **is same as** $\cos ec\theta$**.**

MCQ WORKSHEET-I
CLASS X: CHAPTER - 8
INTRODUCTION TO TRIGONOMETRY

1. In ΔOPQ, right-angled at P, OP = 7 cm and OQ – PQ = 1 cm, then the values of sin Q.
(a) $\frac{7}{25}$ (b) $\frac{24}{25}$ (c) 1 (d) none of the these

2. If sin A = $\frac{24}{25}$, then the value of cosA is
(a) $\frac{7}{25}$ (b) $\frac{24}{25}$ (c) 1 (d) none of the these

3. In ΔABC, right-angled at B, AB = 5 cm and $\angle ACB = 30°$ then the length of the side BC is
(a) $5\sqrt{3}$ (b) $2\sqrt{3}$ (c) 10 cm (d) none of these

4. In ΔABC, right-angled at B, AB = 5 cm and $\angle ACB = 30°$ then the length of the side AC is
(a) $5\sqrt{3}$ (b) $2\sqrt{3}$ (c) 10 cm (d) none of these

5. The value of $\frac{2\tan 30^0}{1+\tan^2 30^0}$ is
(a) sin 60° (b) cos 60° (c) tan 60° (d) sin 30°

6. The value of $\frac{1-\tan^2 45^0}{1+\tan^2 45^0}$ is
(a) tan 90° (b) 1 (c) sin 45° (d) 0

7. sin 2A = 2 sin A is true when A =
(a) 0° (b) 30° (c) 45° (d) 60°

8. The value of $\frac{2\tan 30^0}{1-\tan^2 30^0}$ is
(a) sin 60° (b) cos 60° (c) tan 60° (d) sin 30°

9. $9\sec^2 A - 9\tan^2 A =$
(a) 1 (b) 9 (c) 8 (d) 0

10. $(1 + \tan A + \sec A)(1 + \cot A - \operatorname{cosec} A) =$
(a) 0 (b) 1 (c) 2 (d) –1

11. $(\sec A + \tan A)(1 - \sin A) =$
(a) sec A (b) sin A (c) cosec A (d) cos A

12. $\frac{1+\tan^2 A}{1+\cot^2 A} =$
(a) $\sec^2 A$ (b) –1 (c) $\cot^2 A$ (d) $\tan^2 A$

MCQ WORKSHEET-II
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INTRODUCTION TO TRIGONOMETRY

1. If sin 3A = cos (A – 26°), where 3A is an acute angle, find the value of A.
(a) 29^0 (b) 30^0 (c) 26^0 (d) 36^0

2. If tan 2A = cot (A – 18°), where 2A is an acute angle, find the value of A.
(a) 29^0 (b) 30^0 (c) 26^0 (d) none of these

3. If sec 4A = cosec (A – 20°), where 4A is an acute angle, find the value of A.
(a) 22^0 (b) 25^0 (c) 26^0 (d) none of these

4. The value of tan 48° tan 23° tan 42° tan 67° is
(a) 1 (b) 9 (c) 8 (d) 0

5. If ΔABC is right angled at C, then the value of cos(A + B) is
(a) 0 (b) 1 (c) ½ (d) n.d.

6. The value of the expression $\left[\frac{\sin^2 22^0 + \sin^2 68^0}{\cos^2 22^0 + \cos^2 68^0} + \sin^2 63^0 + \cos 63^0 \sin 27^0\right]$ is
(a) 3 (b) 0 (c) 1 (d) 2

7. If cos A = $\frac{24}{25}$, then the value of sinA is
(a) $\frac{7}{25}$ (b) $\frac{24}{25}$ (c) 1 (d) none of the these

8. If ΔABC is right angled at B, then the value of cos(A + C) is
(a) 0 (b) 1 (c) ½ (d) n.d.

9. If tanA = $\frac{4}{3}$, then the value of cosA is
(a) $\frac{3}{5}$ (b) $\frac{4}{3}$ (c) 1 (d) none of the these

10. If ΔABC is right angled at C, in which AB = 29 units, BC = 21 units and $\angle ABC = \alpha$. Determine the values of $\cos^2\alpha + \sin^2\alpha$ is
(a) 0 (b) 1 (c) ½ (d) n.d.

11. In a right triangle ABC, right-angled at B, if tan A = 1, then the value of 2 sin A cos A =
(a) 0 (b) 1 (c) ½ (d) n.d.

12. Given 15 cot A = 8, then sin A =
(a) $\frac{3}{5}$ (b) $\frac{4}{3}$ (c) 1 (d) none of the these

MCQ WORKSHEET-III
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1. In a triangle PQR, right-angled at Q, PR + QR = 25 cm and PQ = 5 cm, then the value of sin P is
(a) $\frac{7}{25}$ (b) $\frac{24}{25}$ (c) 1 (d) none of the these

2. In a triangle PQR, right-angled at Q, PQ = 3 cm and PR = 6 cm, then $\angle QPR$ =
(a) 0° (b) 30° (c) 45° (d) 60°

3. If sin (A – B) = $\frac{1}{2}$ and cos(A + B) = $\frac{1}{2}$, then the value of A and B, respectively are
(a) 45° and 15° (b) 30° and 15° (c) 45° and 30° (d) none of these

4. If sin (A – B) = 1 and cos(A + B) = 1, then the value of A and B, respectively are
(a) 45° and 15° (b) 30° and 15° (c) 45° and 30° (d) none of these

5. If tan (A – B) = $\frac{1}{\sqrt{3}}$ and tan (A + B) = $\sqrt{3}$, then the value of A and B, respectively are
(a) 45° and 15° (b) 30° and 15° (c) 45° and 30° (d) none of these

6. If cos (A – B) = $\frac{\sqrt{3}}{2}$ and sin (A + B) = 1, then the value of A and B, respectively are
(a) 45° and 15° (b) 30° and 15° (c) 60° and 30° (d) none of these

7. The value of $2\cos^2 60^0 + 3\sin^2 45^0 - 3\sin^2 30^0 + 2\cos^2 90^0$ is
(a) 1 (b) 5 (c) 5/4 (d) none of these

8. sin 2A = 2 sin AcosA is true when A =
(a) 0° (b) 30° (c) 45° (d) any angle

9. sin A = cosA is true when A =
(a) 0° (b) 30° (c) 45° (d) any angle

10. If sinA = $\frac{1}{2}$, then the value of $3\cos A - 4\cos^3 A$ is
(a) 0 (b) 1 (c) ½ (d) n.d.

11. If 3cotA = 4, then the value of $\cos^2 A - \sin^2 A$ is
(a) $\frac{3}{4}$ (b) $\frac{7}{25}$ (c) ½ (d) $\frac{24}{25}$

12. If 3tanA = 4, then the value of $\frac{3\sin A + 2\cos A}{3\sin A - 2\cos A}$ is
(a) 1 (b) $\frac{7}{25}$ (c) 3 (d) $\frac{24}{25}$

MCQ WORKSHEET-IV
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INTRODUCTION TO TRIGONOMETRY

1. Value of θ, for $\sin 2\theta = 1$, *where* $0^0 < \theta < 90^0$ is:
(a) 30° (b) 60° (c) 45° (d) 135^0.

2. Value of $\sec^2 26^0 - \cot^2 64^0$ is:
(a) 1 (b) –1 (c) 0 (d) 2

3. Product $\tan 1^0 . \tan 2^0 . \tan 3^0 \ldots\ldots \tan 89^0$ is:
(a) 1 (b) –1 (c) 0 (d) 90

4. $\sqrt{1+\tan^2\theta}$ is equal to:
(a) $\cot\theta$ (b) $\cos\theta$ (c) $\cos ec\theta$ (d) $\sec\theta$

5. If A + B = 90^0, cot B = $\frac{3}{4}$ then tanA is equal to;
(a) $\frac{3}{4}$ (b) $\frac{4}{3}$ (c) $\frac{1}{4}$ (d) $\frac{1}{3}$

6. Maximum value of $\frac{1}{\cos ec\theta}$, $0^0 < \theta < 90^0$ is:
(a) 1 (b) –1 (c) 2 (d) $\frac{1}{2}$

7. If $\cos\theta = \frac{1}{2}$, $\sin\phi = \frac{1}{2}$ then value of $\theta + \phi$ is
(a) 30° (b) 60° (c) 90° (d) 120^0.

8. If Sin (A + B) = 1 = cos(A – B) then
(a) A = B = 90^0 (b) A = B = 0^0 (c) A = B = 45^0 (d) A = 2B

9. The value of $\sin 60^0 \cos 30^0 - \cos 60^0 \sin 30^0$ is
(a) 1 (b) –1 (c) 0 (d) none of these

10. The value of $2\sin^2 30^0 - 3\cos^2 45^0 + \tan^2 60^0 + 3\sin^2 90^0$ is
(a) 1 (b) 5 (c) 0 (d) none of these

PRACTICE QUESTIONS

CLASS X: CHAPTER - 8

INTRODUCTION TO TRIGONOMETRY

TRIGONOMETRIC RATIOS

1. If $\tan\theta = \frac{1}{\sqrt{5}}$, what is the value of $\frac{\cos ec^2\theta - \sec^2\theta}{\cos ec^2\theta + \sec^2\theta}$?

2. If $\sin\theta = \frac{4}{5}$, find the value of $\frac{\sin\theta\tan\theta - 1}{2\tan^2\theta}$.

3. If cosA = $\frac{1}{2}$, find the value of $\frac{2\sec A}{1+\tan^2 A}$.

4. If $\sin\theta = \frac{\sqrt{3}}{2}$, find the value of all T– ratios of θ.

5. If $\cos\theta = \frac{7}{25}$, find the value of all T– ratios of θ.

6. If $\tan\theta = \frac{15}{8}$, find the value of all T– ratios of θ.

7. If $\cot\theta = 2$, find the value of all T– ratios of θ.

8. If $\text{cosec}\,\theta = \sqrt{10}$, find the value of all T– ratios of θ.

9. If $\tan\theta = \frac{4}{3}$, show that $(\sin\theta + \cos\theta) = \frac{7}{5}$.

10. If $\sec\theta = \frac{5}{4}$, show that $\frac{(\sin\theta - 2\cos\theta)}{(\tan\theta - \cot\theta)} = \frac{12}{7}$.

11. If $\tan\theta = \frac{1}{\sqrt{7}}$, show that $\frac{(\cos ec^2\theta - \sec^2\theta)}{(\cos ec^2\theta + \sec^2\theta)} = \frac{3}{4}$.

12. If $\cos ec\theta = 2$, show that $\left\{\cot\theta + \frac{\sin\theta}{1+\cos\theta}\right\} = 2$.

13. If $\sec\theta = \frac{5}{4}$, verify that $\frac{\tan\theta}{(1+\tan^2\theta)} = \frac{\sin\theta}{\sec\theta}$.

14. If $\cos\theta = 0.6$, show that $(5\sin\theta - 3\tan\theta) = 0$.

15. In a triangle ACB, right-angled at C, in which AB = 29 units, BC = 21 units and $\angle ABC = \theta$. Determine the values of (i) $\cos^2\theta + \sin^2\theta$ (ii) $\cos^2\theta - \sin^2\theta$

16. In a triangle ABC, right-angled at B, in which AB = 12 cm and BC = 5cm. Find the value of cosA, cosecA, cosC and cosecC.

17. In a triangle ABC, $\angle B = 90^0$, AB = 24 cm and BC = 7 cm. Find (i) SinA, CosA (ii) SinC, CosC.

PRACTICE QUESTIONS

CLASS X: CHAPTER - 8

INTRODUCTION TO TRIGONOMETRY

T – RATIOS OF SOME PARTICULAR ANGLES

Evaluate each of the following:

1. $\sin 60^0 \cos 30^0 + \cos 60^0 \sin 30^0$

2. $\cos 60^0 \cos 30^0 - \sin 60^0 \sin 30^0$

3. $\cos 45^0 \cos 30^0 + \sin 45^0 \sin 30^0$

4. $\sin 60^0 \sin 45^0 - \cos 60^0 \cos 45^0$

5. $\dfrac{\sin 30^0}{\cos 45^0} + \dfrac{\cot 45^0}{\sec 60^0} - \dfrac{\sin 60^0}{\tan 45^0} - \dfrac{\cos 30^0}{\sin 90^0}$

6. $\dfrac{\tan^2 60^0 + 4\cos^2 45^0 + 3\cos ec^2 60^0 + 2\cos^2 90^0}{2\cos ec 30^0 + 3\sec 60^0 - \dfrac{7}{3}\cot^2 30^0}$

7. $4(\sin^4 30^0 + \cos^4 60^0) - 3(\cos^2 45^0 - \sin^2 90^0) + 5\cos^2 90^0$

8. $\dfrac{4}{\cot^2 30^0} + \dfrac{1}{\sin^2 30^0} - 2\cos^2 45^0 - \sin^2 0^0$

9. $\dfrac{1}{\cos^2 30^0} + \dfrac{1}{\sin^3 30^0} - \dfrac{1}{2}\tan^2 45^0 - 8\sin^2 90^0$

10. $\cot^2 30^0 - 2\cos^2 30^0 - \dfrac{3}{4}\sec^2 45^0 + \dfrac{1}{4}\cos ec^2 30^0$

11. $(\sin^2 30^0 + 4\cot^2 45^0 - \sec^2 60^0)(\cos ec^2 45^0 \sec^2 30^0)$

12. In right triangle ABC, $\angle B = 90^0$, AB = 3cm and AC = 6cm. Find $\angle C$ and $\angle A$.

13. If $A = 30^0$, verify that:

(i) $\sin 2A = \dfrac{2\tan A}{1+\tan^2 A}$ (ii) $\cos 2A = \dfrac{1-\tan^2 A}{1+\tan^2 A}$ (iii) $\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$

14. If $A = 45^0$, verify that

(i) $\sin 2A = 2\sin A \cos A$ (ii) $\cos 2A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$

15. Using the formula, $\cos A = \sqrt{\dfrac{1+\cos 2A}{2}}$, find the value of $\cos 30^0$, it being given that $\cos 60^0 = \dfrac{1}{2}$

16. Using the formula, $\sin A = \sqrt{\dfrac{1-\cos 2A}{2}}$, find the value of $\sin 30^0$, it being given that $\cos 60^0 = \dfrac{1}{2}$

17. Using the formula, $\tan 2A = \dfrac{2\tan A}{1-\tan^2 A}$, find the value of $\tan 60^0$, it being given that $\tan 30^0 = \dfrac{1}{\sqrt{3}}$.

18. If $\sin(A - B) = \dfrac{1}{2}$ and $\cos(A + B) = \dfrac{1}{2}$, then find the value of A and B.

19. If $\sin(A + B) = 1$ and $\cos(A - B) = 1$, then find the value of A and B.

20. If $\tan(A - B) = \dfrac{1}{\sqrt{3}}$ and $\tan(A + B) = \sqrt{3}$, then find the value of A and B.

21. If $\cos(A - B) = \dfrac{\sqrt{3}}{2}$ and $\sin(A + B) = 1$, then find the value of A and B.

22. If A and B are acute angles such that $\tan A = \dfrac{1}{3}$, $\tan B = \dfrac{1}{2}$ and $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$, show that $A + B = 45^0$.

23. If $A = B = 45^0$, verify that:

a) $\sin(A + B) = \sin A\cos B + \cos A\sin B$
b) $\sin(A - B) = \sin A\cos B - \cos A\sin B$
c) $\cos(A + B) = \cos A\cos B - \sin A\sin B$
d) $\cos(A - B) = \cos A\cos B + \sin A\sin B$
e) $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$
f) $\tan(A-B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$

24. If $A = 60^0$ and $B = 30^0$, verify that:

a) $\sin(A + B) = \sin A\cos B + \cos A\sin B$
b) $\sin(A - B) = \sin A\cos B - \cos A\sin B$
c) $\cos(A + B) = \cos A\cos B - \sin A\sin B$
d) $\cos(A - B) = \cos A\cos B + \sin A\sin B$
e) $\tan(A+B) = \dfrac{\tan A + \tan B}{1 - \tan A \tan B}$
f) $\tan(A-B) = \dfrac{\tan A - \tan B}{1 + \tan A \tan B}$

25. Evaluate:

$$\frac{\sin^2 45^0 + \dfrac{3}{4}\cos ec^2 30^0 - \cos 60^0 + \tan^2 60^0}{\sin^2 30^0 + \cos^2 60^0 + \dfrac{1}{2}\sec^2 45^0}$$

PRACTICE QUESTIONS

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INTRODUCTION TO TRIGONOMETRY

T – RATIOS OF COMPLEMENTARY ANGLES

1. Evaluate: $\cot\theta \tan(90^0 - \theta) - \sec(90^0 - \theta)\,\text{cosec}\theta + (\sin^2 25^0 + \sin^2 65^0) + \sqrt{3}(\tan 5^0 . \tan 15^0 . \tan 30^0 . \tan 75^0 . \tan 85^0)$.

2. Evaluate without using tables: $\dfrac{\sec\theta \cos ec(90^0 - \theta) - \tan\theta \cot(90^0 - \theta) + (\sin^2 35^0 + \sin^2 55^0)}{\tan 10^0 \tan 20^0 \tan 45^0 \tan 70^0 \tan 80^0}$

3. Evaluate: $\dfrac{\sec^2 54^0 - \cot^2 36^0}{\cos ec^2 57^0 - \tan^2 33^0} + 2\sin^2 38^0 \sec^2 52^0 - \sin^2 45^0$.

4. Express $\sin 67^0 + \cos 75^0$ in terms of trigonometric ratios of angles between 0^0 and 45^0.

5. If $\sin 4A = \cos(A - 20^0)$, where A is an acute angle, find the value of A.

6. If A, B and C are the interior angles of triangle ABC, prove that $\tan\left(\dfrac{B+C}{2}\right) = \cot\dfrac{A}{2}$

7. If A, B, C are interior angles of a ΔABC, then show that $\cos\left(\dfrac{B+C}{2}\right) = \sin\dfrac{A}{2}$.

8. If A, B, C are interior angles of a ΔABC, then show that $\cos ec\left(\dfrac{A+C}{2}\right) = \sec\dfrac{B}{2}$.

9. If A, B, C are interior angles of a ΔABC, then show that $\cot\left(\dfrac{B+A}{2}\right) = \tan\dfrac{C}{2}$.

10. Without using trigonometric tables, find the value of $\dfrac{\cos 70^0}{\sin 20^0} + \cos 57^0 \cos ec 33^0 - 2\cos 60^0$.

11. If $\sec 4A = \text{cosec}(A - 20^0)$, where 4A is an acute angle, find the value of A.

12. If $\tan 2A = \cot(A - 40^0)$, where 2A is an acute angle, find the value of A.

13. Evaluate $\tan 10^0 \tan 15^0 \tan 75^0 \tan 80^0$

14. Evaluate: $\left[\dfrac{\sin^2 22^0 + \sin^2 68^0}{\cos^2 22^0 + \cos^2 68^0} + \sin^2 63^0 + \cos 63^0 \sin 27^0\right]$

15. Express $\tan 60^0 + \cos 46^0$ in terms of trigonometric ratios of angles between 0^0 and 45^0.

16. Express $\sec 51^0 + \text{cosec}\, 25^0$ in terms of trigonometric ratios of angles between 0^0 and 45^0.

17. Express $\cot 77^0 + \sin 54^0$ in terms of trigonometric ratios of angles between 0^0 and 45^0.

18. If $\tan 3A = \cot(3A - 60^0)$, where 3A is an acute angle, find the value of A.

19. If $\sin 2A = \cos(A + 36^0)$, where 2A is an acute angle, find the value of A.

20. If $\text{cosec}A = \sec(A - 10^0)$, where A is an acute angle, find the value of A.

21. If $\sin 5\theta = \cos 4\theta$, where 5θ and 4θ are acute angles, find the value of θ.

22. If $\tan 2A = \cot (A - 18^0)$, where $2A$ is an acute angle, find the value of A.

23. If $\tan 2\theta = \cot(\theta + 6^0)$, where 2θ and $\theta + 6^0$ are acute angles, find the value of θ.

24. Evaluate:

$$\frac{2\sin 68^0}{\cos 22^0} - \frac{2\cot 15^0}{5\tan 75^0} - \frac{3\tan 45^0 \tan 20^0 \tan 40^0 \tan 50^0 \tan 70^0}{5}$$

25. Evaluate:

$$\frac{\cos(90^0 - \theta)\sec(90^0 - \theta)\tan\theta}{\cos ec(90^0 - \theta)\sin(90^0 - \theta)\cot(90^0 - \theta)} + \frac{\tan(90^0 - \theta)}{\cot\theta} + 2$$

26. Evaluate:

$$\frac{\sin 18^0}{\cos 72^0} + \sqrt{3}\left\{\tan 10^0 \tan 30^0 \tan 40^0 \tan 50^0 \tan 80^0\right\}$$

27. Evaluate:

$$\frac{3\cos 55^0}{7\sin 35^0} - \frac{4(\cos 70^0 \cos ec 20^0)}{7(\tan 5^0 \tan 25^0 \tan 45^0 \tan 65^0 \tan 85^0)}$$

28. Evaluate:

$$\cos(40^0 - \theta) - \sin(50^0 + \theta) + \frac{\cos^2 40^0 + \cos^2 50^0}{\sin^2 40^0 + \sin^2 50^0}$$

29. If $A + B = 90^0$, prove that $\sqrt{\frac{\tan A \tan B + \tan A \cot B}{\sin A \sec B} - \frac{\sin^2 B}{\cos^2 A}} = \tan A$

30. If $\cos 2\theta = \sin 4\theta$, where 2θ and 4θ are acute angles, find the value of θ.

PRACTICE QUESTIONS

CLASS X: CHAPTER - 8

INTRODUCTION TO TRIGONOMETRY

TRIGONOMETRIC IDENTITIES

1. Prove that $\frac{\cos\theta}{1-\tan\theta}+\frac{\sin\theta}{1-\cot\theta}=\sin\theta+\cos\theta$.

2. Prove that $\frac{1}{2}\left\{\frac{\sin\theta}{1+\cos\theta}+\frac{1+\cos\theta}{\sin\theta}\right\}=\frac{1}{\sin\theta}$.

3. Prove that: $\frac{\tan^3\alpha}{1+\tan^2\alpha}+\frac{\cot^3\alpha}{1+\cot^2\alpha}=\sec\alpha\,\mathrm{cosec}\alpha-2\sin\alpha\cos\alpha$

4. Prove that: $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}=1+\tan A+\cot A=1+\sec A\,\mathrm{cosec}A$.

5. Prove that:. $\frac{1+\sin A-\cos A}{1+\sin A+\cos A}=\sqrt{\frac{1-\cos A}{1+\cos A}}$

6. Prove that $(\tan A+\mathrm{cosec}B)^2-(\cot B-\sec A)^2=2\tan A\cot B(\mathrm{cosec}A+\sec B)$.

7. Prove that: $\frac{\cos A}{1+\sin A}+\frac{1+\sin A}{\cos A}=2\sec A$.

8. Prove that: $\frac{\cos A-\sin A+1}{\cos A+\sin A-1}=\mathrm{cosec}A+\cot A$.

9. Prove that: $\frac{\sin A+\cos A}{\sin A-\cos A}+\frac{\sin A-\cos A}{\sin A+\cos A}=\frac{2}{\sin^2 A-\cos^2 A}=\frac{2}{2\sin^2 A-1}=\frac{2}{1-2\cos^2 A}$.

10. Prove that $\frac{\sin A}{\cot A+\mathrm{cosec}A}=2+\frac{\sin A}{\cot A-\mathrm{cosec}A}$.

11. Prove that $\sqrt{\frac{\sec A-1}{\sec A+1}}+\sqrt{\frac{\sec A+1}{\sec A-1}}=2\,\mathrm{cosec}A$.

12. Prove that: $\frac{1}{\mathrm{cosec}A-\cot A}-\frac{1}{\sin A}=\frac{1}{\sin A}-\frac{1}{\mathrm{cosec}A+\cot A}$.

13. Prove that: $\frac{\tan\theta+\sec\theta-1}{\tan\theta-\sec\theta+1}=\sec\theta+\tan\theta=\frac{1+\sin\theta}{\cos\theta}$

14. If x = $a\sin\theta+b\cos\theta$ and y = $a\cos\theta+b\sin\theta$, prove that $x^2+y^2=a^2+b^2$.

15. If $\sec\theta+\tan\theta=m$, show that $\left(\frac{m^2-1}{m^2+1}\right)=\sin\theta$.

16. Prove that: $\frac{1+\cos\theta+\sin\theta}{1+\cos\theta-\sin\theta}=\frac{1+\sin\theta}{\cos\theta}$.

17. Prove that $\sec^4 A(1-\sin^4 A)-2\tan^2 A=1$

18. If $\operatorname{cosec}\theta-\sin\theta=m$ and $\sec\theta-\cos\theta=n$, prove that $\left(m^2n\right)^{2/3}+\left(mn^2\right)^{2/3}=1$

19. If $\tan\theta+\sin\theta=m$ and $\tan\theta-\sin\theta=n$, show that $m^2-n^2=4\sqrt{mn}$

20. If $a\cos\theta-b\sin\theta=c$, prove that $\left(a\sin\theta+b\cos\theta\right)=\pm\sqrt{a^2+b^2-c^2}$

21. If $\cos\theta+\sin\theta=\sqrt{2}\cos\theta$, prove that $\cos\theta-\sin\theta=\sqrt{2}\sin\theta$

22. If $\left(\frac{x}{a}\sin\theta-\frac{y}{b}\cos\theta\right)=1$ and $\left(\frac{x}{a}\cos\theta+\frac{y}{b}\sin\theta\right)=1$, prove that $\left(\frac{x^2}{a^2}+\frac{y^2}{b^2}\right)=2$

23. If $\left(\tan\theta+\sin\theta\right)=m$ and $\left(\tan\theta-\sin\theta\right)=n$ prove that $\left(m^2-n^2\right)^2=16mn$

24. If $\operatorname{cosec}\theta-\sin\theta=a^3$ and $\operatorname{sec}\theta-\cos\theta=b^3$, prove that $a^2b^2(a^2+b^2)=1$

25. If $a\cos^3\theta+3a\sin^2\theta\cos\theta=m$ and $a\sin^3\theta+3a\sin\theta\cos^2\theta=n$, prove that $\left(m+n\right)^{2/3}+(m-n)^{2/3}=2a^{2/3}$

26. Prove that $\sqrt{\sec^2\theta+\operatorname{cosec}^2\theta}=\tan\theta+\cot\theta$.

27. Prove the identity: $\frac{\sin\theta-\cos\theta+1}{\sin\theta+\cos\theta-1}=\frac{1}{\sec\theta-\tan\theta}$.

28. Prove the identity: $\sec^6\theta=\tan^6\theta+3\tan^2\theta.\sec^2\theta+1$.

29. Prove the identity: $(\sin A+\operatorname{cosec} A)^2+(\cos A+\sec A)^2=7+\tan^2 A+\cot^2 A$.

30. If $x\sin^3\theta+y\cos^3\theta=\sin\theta\cos\theta$ and $x\sin\theta=y\cos\theta$, prove that $x^2+y^2=1$.

31. If $\sec\theta=x+\frac{1}{4x}$, Prove that $\sec\theta+\tan\theta=2x$ or $\frac{1}{2x}$.

32. Prove that $\left(1+\frac{1}{\tan^2 A}\right)\left(1+\frac{1}{\cot^2 A}\right)=\frac{1}{\sin^2 A-\sin^4 A}$.

33. If $\cot\theta+\tan\theta=x$ and $\sec\theta-\cos\theta=y$, prove that $\left(x^2y\right)^{2/3}-\left(xy^2\right)^{2/3}=1$.

34. If $\frac{\cos\alpha}{\cos\beta}=m$ *and* $\frac{\cos\alpha}{\sin\beta}=n$, show that $(m^2+n^2)\cos^2\beta=n^2$.

35. If $\operatorname{cosec}\theta-\sin\theta=a$ and $\sec\theta-\cos\theta=b$, prove that $a^2b^2(a^2+b^2+3)=1$

36. If $x = r\sin A\cos C$, $y = r\sin A\sin C$ and $z = r\cos A$, prove that $r^2 = x^2 + y^2 + z^2$.

37. If $\tan A = n \tan B$ and $\sin A = m\sin B$, prove that $\cos^2 A = \frac{m^2 - 1}{n^2 - 1}$.

38. If $\sin\theta + \sin^2\theta = 1$, find the value of $\cos^{12}\theta + 3\cos^{10}\theta + 3\cos^8\theta + \cos^6\theta + 2\cos^4\theta + 2\cos^2\theta - 2$..

39. Prove that: $(1 - \sin\theta + \cos\theta)^2 = 2(1 + \cos\theta)(1 - \sin\theta)$

40. If $\sin + \sin^2\theta = 1$, prove that $\cos^2\theta + \cos^4\theta = 1$.

41. If $a\sec\theta + b\tan\theta + c = 0$ and $p\sec\theta + q\tan\theta + r = 0$, prove that $(br - qc)^2 - (pc - ar)^2 = (aq - bp)^2$.

42. If $\sin\theta + \sin^2\theta + \sin^3\theta = 1$, then prove that $\cos^6\theta - 4\cos^4\theta + 8\cos^2\theta = 4$.

43. If $\tan^2\theta = 1 - a^2$, prove that $\sec\theta + \tan^3\theta\operatorname{cosec}\theta = (2 - a^2)^{3/2}$.

44. If $x = a\sec\theta + b\tan\theta$ and $y = a\tan\theta + b\sec\theta$, prove that $x^2 - y^2 = a^2 - b^2$.

45. If $3\sin\theta + 5\cos\theta = 5$, prove that $5\sin\theta - 3\cos\theta = \pm 3$.